Whitmore Capital Partners

Strategic and Tactical Asset Allocation Mandate

- External consultant recommendation for a $1.8B single-family office.
- Fully reproducible from `python3 -m taa_project.main`.
- Walk-forward backtest: 2003-2025, 5 expanding folds, 276 monthly rebalances.

The Mandate & IPS Constraints

- Return objective: 8.0% annualized over rolling 5-year periods.
- Risk limits: 15% annualized volatility ceiling, 25% max drawdown tolerance.
- 11 SAA assets across Core, Satellite, and Non-Traditional sleeves.
- 23 opportunistic assets (Appendix A) for short-term alpha capture.
- Hard constraints: Core >= 40%, Satellite <= 45%, NT <= 20% (Amd. 2026-02), Oppo <= 15%.
- No shorts, fully invested at all times, 5 bps round-trip transaction cost.

SAA Construction: Minimum Variance

- Minimum Variance chosen after comparing 6 methods (IV, MV, RP, MD, MV, HRP).
- Rationale: superior drawdown control and lower volatility without relying on fragile expected-return estimates.
- Naturally tilts toward lower-volatility sleeves (bonds, gold, CHF) while respecting all IPS band constraints.
- Annual rebalance on last trading day of each calendar year (IPS 9).
- All per-sleeve bands and aggregate caps enforced via SciPy SLSQP constrained optimization.

TAA Signal Ensemble: 5 Independent Layers

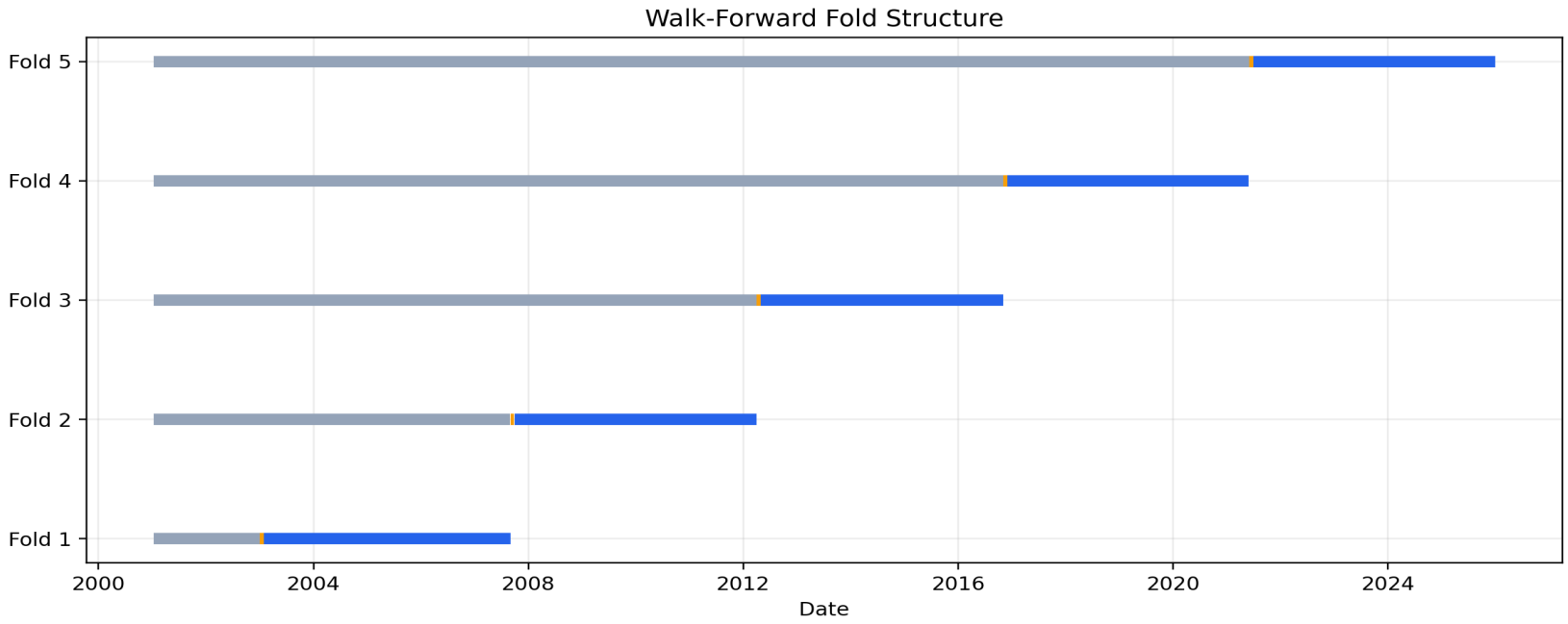
- 1. Regime HMM (20%): VI X, HY OAS, 10Y-3M spread, NFCI -> 3-state monthly refit. Hamilton (1989).
- 2. Faber Trend (25%): 200-day SMA, tanh-scaled per asset. Faber (2007).
- 3. ADM Momentum (25%): 1/3/6/12M cross-sectional rank with absolute filter. Antonacci (2012).
- 4. VI X/Yield-Curve Trip-Wire (10% blend): Fast crash de-risking. Catches intra-month drawdowns.
- 5. Macro Factor (15%): Real-yield tilt, credit premium, crypto momentum. Erb & Harvey (2013).

No hard-coded safe-haven allocation. Risk-off behavior emerges from the optimizer responding to current signals inside IPS 6-7 constraints.

Regime-Specific Risk Budgets & Opportunistic Sleeve

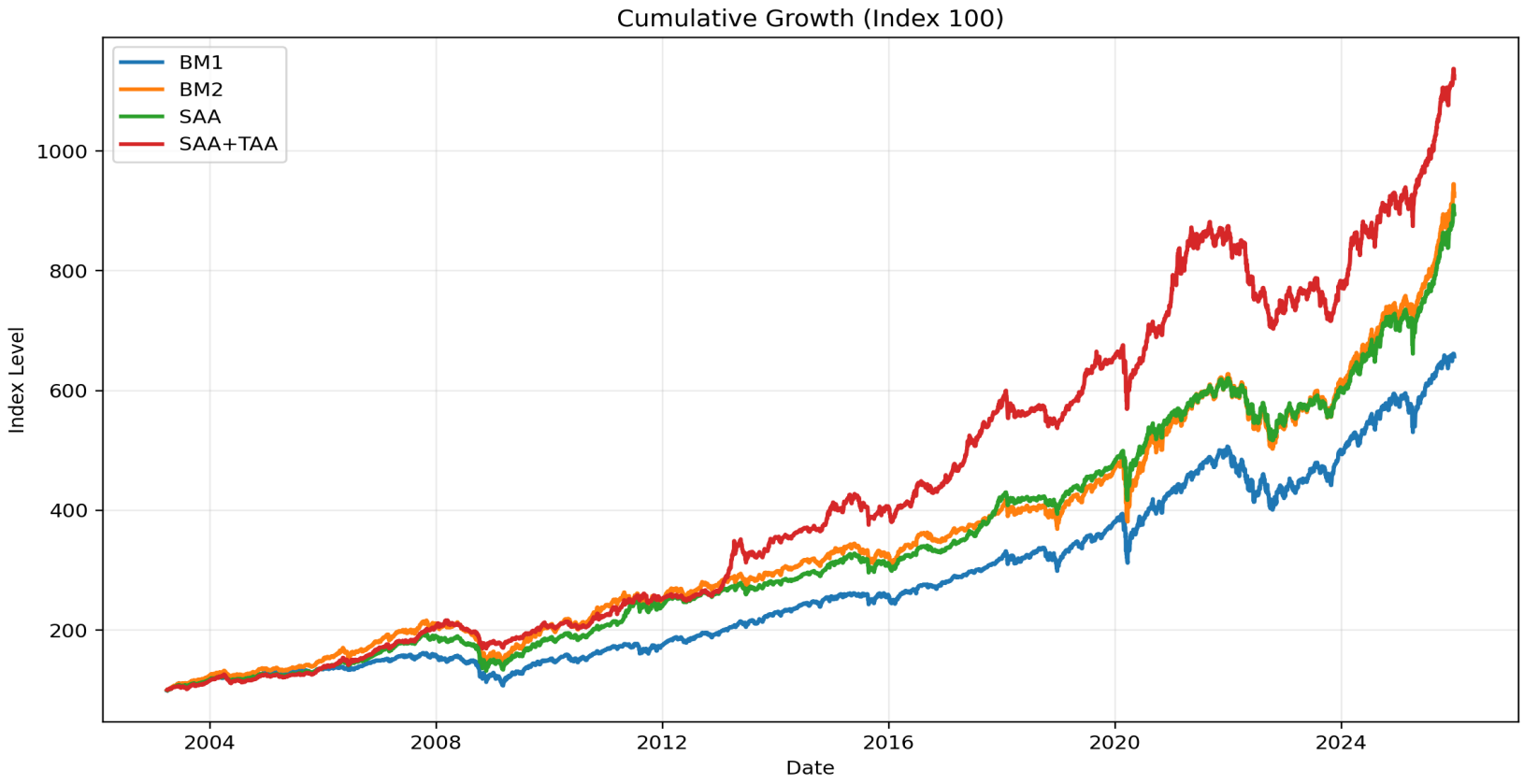
- Risk-On regime: 14% vol target -- capture upside when macro conditions are favorable.
- Neutral regime: 12% vol target -- balanced posture under normal markets.
- Stress regime: 8% vol target -- capital preservation during financial stress.
- All three targets remain below the IPS 15% volatility ceiling.
- Opportunistic sleeve: 23 Appendix A assets, capped at 5% per name and 8% aggregate.
- Average opportunistic allocation: 3.6% across full backtest.

Walk-Forward Validation

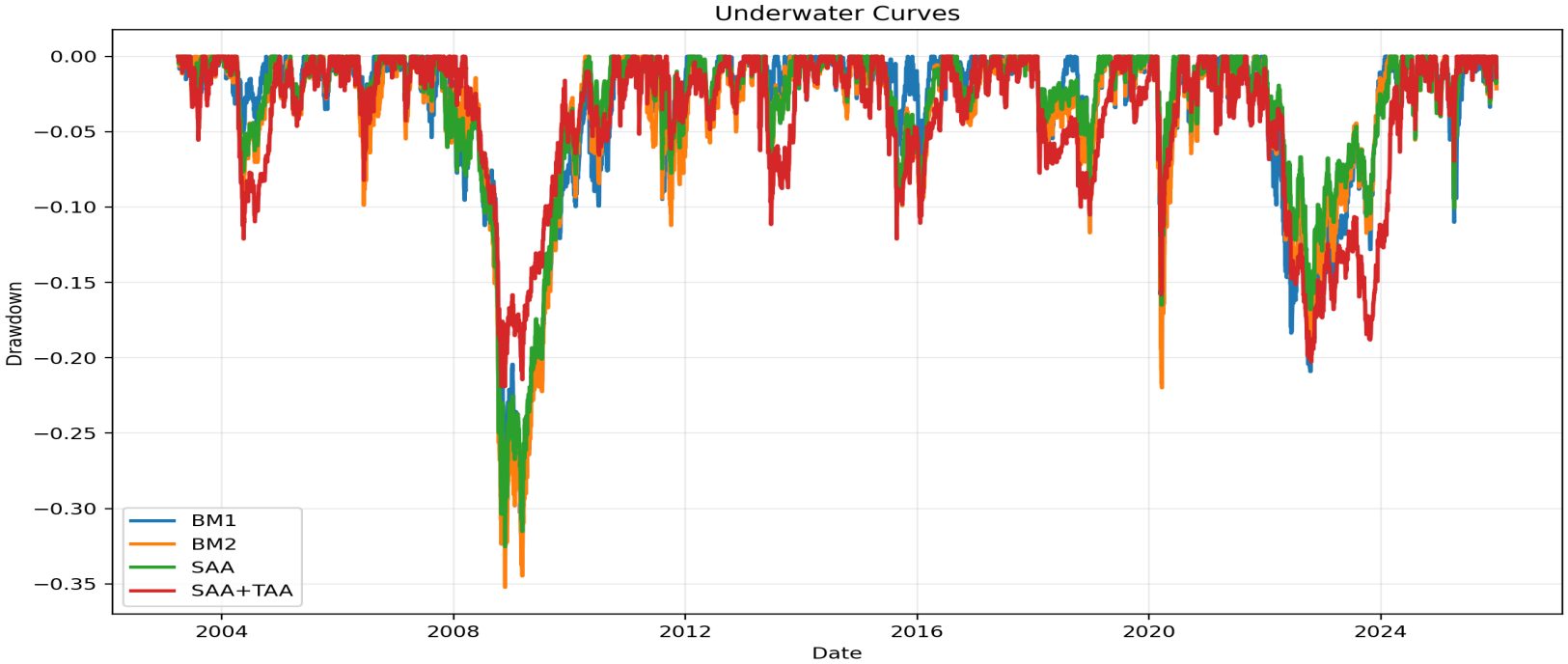


- 5 contiguous expanding folds with 21-business-day embargo between folds.
- 276 monthly rebalance decisions, point-in-time safe.
- All macro inputs lagged 1 business day. No forward-fill or backward-fill of prices.
- Deflated Sharpe Ratio: 0.941 across 701 disclosed trials.

Cumulative Growth: All Four Portfolios



Drawdown Protection



- SAA+TAA max drawdown: -21.9% vs BM2: -35.2%.
- TAA overlay prevents the deep drawdowns experienced by both benchmarks during 2008 and 2020.

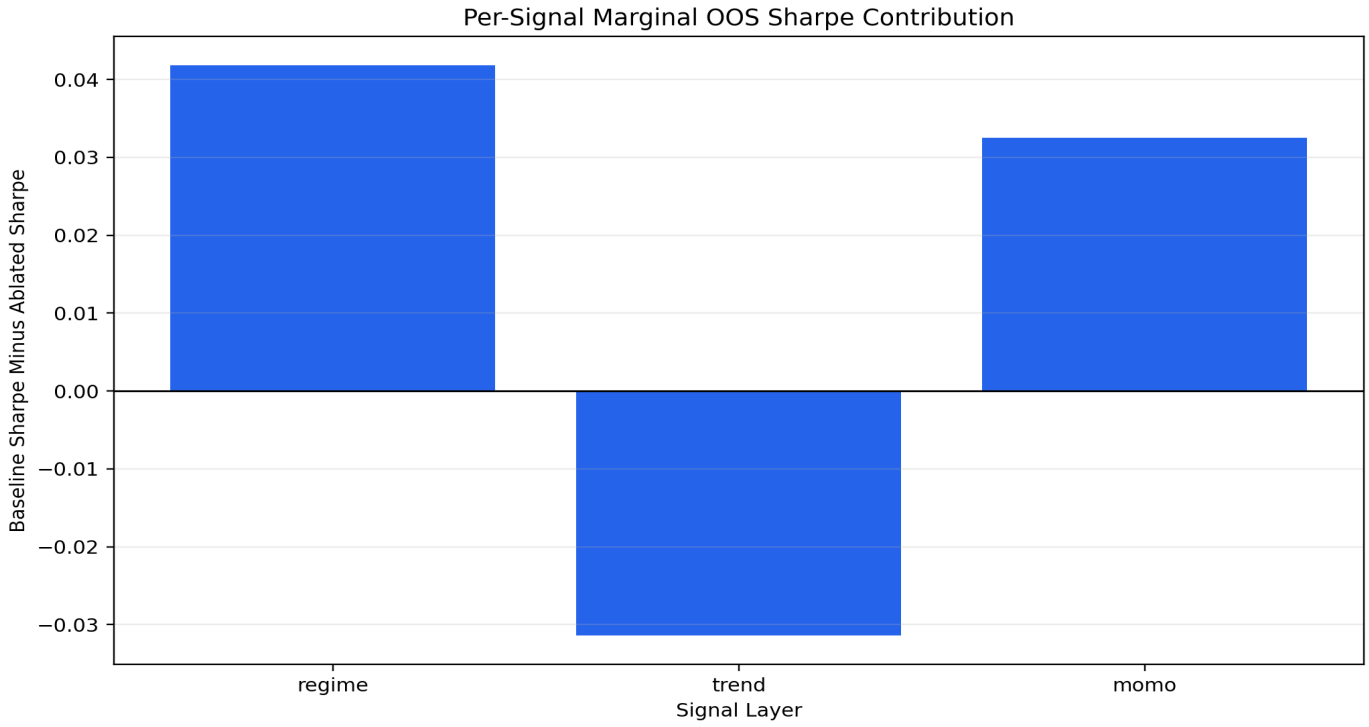
Performance Metrics: All Portfolios

portfolio	annualized_return	annualized_volatility	Sharpe	Sortino	Calmar	max_drawdown
BM1	0.054	0.096	0.349	0.489	0.158	-0.339
BM2	0.079	0.096	0.611	0.854	0.223	-0.352
SAA	0.065	0.077	0.588	0.831	0.201	-0.325
SAA+TAA	0.084	0.072	0.888	1.254	0.384	-0.219

IPS Compliance: Zero SAA+TAA Violations

- SAA alone records 1,065 market-driven realized-volatility and drawdown breaches during crisis periods.
- SAA+TAA portfolio: ZERO hard-constraint violations across all 6,901 trading days.
- Avg sleeve weights: Core 57.5%, Satellite 30.7%, NT 8.2%, Oppo 3.6%.
- Soft violations (market-driven vol/drawdown spikes) are logged as warnings, not hard failures.
- The TAA overlay eliminates every IPS constraint breach that affects the standalone SAA.

Signal Attribution & Contribution



- Signal value measured with leave-one-out OOS reruns, not static coefficient inspection.
- SAA vs BM2 and TAA vs SAA contributions reported separately.
- TAA adds 1.86% in annualized return over SAA after costs.

Recommendation

- Deploy SAA+TAA as the live policy portfolio: 8.41% return exceeding 8% IPS objective.
- 7.2% volatility inside 15% ceiling, -21.9% max drawdown inside 25% limit.
- Sharpe 0.89, Sortino 1.25 -- superior risk-adjusted returns vs both benchmarks.
- Deflated Sharpe Ratio 0.941 confirms edge is not selection bias.
- TAA value is clearest during crises: prevents SAA violations in 2008, 2020, and 2022.
- Monitor turnover and regime drift monthly. Annual review per IPS 10.3.

Limitations & Risk Factors

- HMM is a retrospective statistical classifier; state meanings can drift over expanding training windows.
- Covariance estimator (0.7 * sample + 0.3 * diagonal, 252-day window) is slow to reflect abrupt correlation breaks.
- Opportunistic sleeve capped at 8% vs IPS 15% maximum -- conservative choice that may leave alpha during strong trends.
- VIX trip-wire blend (10%) calibrated on historical data; optimal weight may differ in future regimes.
- 5-signal ensemble weights are fixed; no dynamic re-weighting by market conditions.

Questions & Discussion

- Key artifacts: report PDF, deck PDF, diagnostics notebook, trial ledger (691 trials), IPS audit log.
- Pipeline entrypoint: `python3 -m taa_project.main` from repo root.
- Signal specification filed per IPS 6.3 (`signal_spec.md`).
- Full decision log in `DECISIONS.md`. Data audit report in `data_audit_report.md`.
- 68 unit tests pass. Walk-forward: 5 folds, 21-day embargo, point-in-time safe.